BMGT347: Final Project Guidelines

Purpose of the Final Project

The project is designed to integrate financial data, quantitative methods, economic reasoning, AI-assisted research workflows, and reproducible coding. The objective is not merely to produce a result, but to build and document a professional research workflow that another analyst could understand, audit, and reproduce.

1. Project Logistics and Milestones

Team size	4-5 students
Team lead	One designated team lead responsible for coordination, communications, and workflow integration
Project proposal deadline	11/01/2026
Final project presentation	12/08/2026, 12/10/2026
Final written report due	12/21/2026

Each team is responsible for establishing a clear division of labor. The team lead should facilitate communication and integration, but all team members are expected to understand the core data, methodology, and results well enough to explain them during the final presentation. By the end of semester, each team member will anonymously evaluate other team members' contribution.

2. Project Requirements

- Use actual financial or economic data from public sources, APIs, or WRDS databases. Appropriate examples include CRSP, Compustat, IBES, OptionMetrics, Fama-French Data Library, Yahoo Finance, SEC EDGAR, FRED, U.S. Treasury data, and other data sources your team can access.
- Use quantitative/statistical methods appropriate to the research question. The methodology should be explained, justified, and implemented correctly.
- Translate statistical results into economic and financial interpretation. A project should explain how the findings might be used by an investor, asset manager, risk manager, corporate finance professional, or market analyst, etc.
- Use AI tools substantively in the research workflow. AI may be used for vibe coding, debugging, workflow design, data pipelines, testing, visualization, documentation, or agent architecture. Examples include Codex, ChatGPT, Claude Code/Claude Work, GitHub Copilot, Cursor, or comparable tools.
- Maintain a GitHub repository for version control and final storage of scripts, documentation, outputs, figures, tables, and reproducibility instructions.
- Include a detailed AI Workflow Appendix showing how AI was used and how AI-generated work was checked and validated.

Core principle for AI use

You may delegate work to AI, but you cannot delegate responsibility for correctness. The team must test, validate, and explain any AI-generated code, results, or interpretations used in the final submission.

3. Required Deliverables

The final report should be organized around the following six substantive components:

1. Problem to solve: Clearly state the financial question, motivation, and why it matters.
2. Data collection and manipulation: Describe data sources, variables, sample period, cleaning, merging, transformations, missing-data treatment, and any important filters.
3. Statistical methodology: Explain the econometric, statistical, machine-learning, portfolio, or valuation methods used. Include benchmark models where appropriate and applicable.
4. Results and economic interpretation: Present the main empirical results using well-designed tables and figures, and explain the economic meaning of the findings.
5. Real-world relevance and application: Explain how the results could be used in an actual investment, risk-management, forecasting, or corporate-finance setting. Discuss implementation constraints.
6. Limitations and future exploration: Identify methodological weaknesses, data limitations, possible biases, and realistic next steps for improving the analysis.

4. Project Proposal Requirements

The proposal should be concise but specific enough to demonstrate that the team has a viable research design. A recommended length is approximately 1-2 pages, excluding references.

- Project title and team members (emails), with the designated team lead clearly identified.
- Primary research question and financial motivation.
- Planned data sources, sample period, and main variables.
- Proposed methodology and benchmark models.
- Expected outputs: tables, figures, backtests, forecasts, dashboard, or other deliverables.
- Planned AI workflow: which tools will be used and for what tasks.
- Planned GitHub repository structure and division of labor.
- Team agreed work plans and milestone dates.

5. Final Presentation and Written Report

The final presentation should focus on the research question, methodology, main results, economic interpretation, and what the team learned from using AI in a quantitative research workflow. Do not spend most of the presentation describing code line by line.

The written report should be professional and concise. Use tables and figures to communicate results, and move lengthy implementation details, additional robustness checks, prompt logs, and technical workflow documentation to appendices where appropriate.

All team members should be prepared to answer questions about the data, methodology, results, code, and AI workflow. The instructor may ask any member of the group to explain a core part of the project.

6. Suggested Ideas for Final Projects

Teams are strongly encouraged to pursue your own research ideas with instructor's permission. Ideally, what your team does in this final project can help your future career. Additionally, teams may refer to the following suggested project ideas.

Important: members should meet and discuss your project as soon as possible. Don't wait until the last minute to do work! Start with a prototype in a simple way and then add enhanced features over time.

Project 1 - Cross-Sectional Quantitative Equity Strategies

Research question: Can well-known quantitative signals predict the cross section of future U.S. stock returns, and can the signals be implemented as realistic portfolio strategies?

Possible strategy families include value, momentum, quality, profitability, investment, size, low volatility, earnings revisions, and composite multifactor scores.

Recommended analysis:

- Construct characteristics at each portfolio formation date and avoid look-ahead bias.
- Rank stocks into quantiles and form long-only or long-short portfolios.
- Measure returns, volatility, Sharpe ratios, drawdowns, turnover, and transaction-cost sensitivity.
- Run factor regressions such as CAPM, Fama-French 3-factor, 5-factor, and momentum models.
- Compare performance across subperiods and market environments.
- For an advanced extension, use machine learning or optimization methods to combine multiple signals rather than assigning arbitrary weights.

Potential data: CRSP, Compustat, CRSP/Compustat merged database, IBES, or a smaller public-data universe such as current and historical S&P 500 constituents.

AI opportunities: Build an AI-assisted research pipeline for data cleaning, signal construction, portfolio formation, backtesting, factor attribution, validation checks, and automated reporting.

Project 2 - Time-Series Market Return Prediction

Research question: Can information known at time t help predict future equity-market returns, direction, or trend?

Possible predictors include technical indicators, valuation variables, term and credit spreads, volatility measures, sentiment variables, market breadth, and other macro-financial indicators.

Recommended analysis:

- Define a forecasting target such as next-week SPY return, next-week positive/negative return, or weekly direction of return upward/downward trend.
- Establish a simple benchmark model before using more complex methods.
- Compare methods such as linear regression, logistic regression, LASSO/Elastic Net, random forest, gradient boosting, or neural networks.
- Use rolling-window or expanding-window out-of-sample validation; random train/test splitting is generally inappropriate for time-series forecasting.
- Explicitly test for look-ahead bias, data leakage, overfitting, and unstable parameter estimates.
- Translate forecasts into an investment rule and compare economic performance with a passive benchmark.

Advanced extension: Macro and market regime modeling

More advanced teams may classify macro-financial regimes - for example expansion, slowdown, inflationary pressure, recession risk, or risk-on/risk-off states - and study whether predictive relationships differ across regimes. A regime model may be built using macro variables, market variables, clustering, Markov-switching models, hidden Markov models, or other well-motivated approaches. The key question is whether regime information improves out-of-sample forecasting or market-timing decisions.

AI opportunities: Use AI to design and test a forecasting pipeline, generate candidate predictors, automate feature engineering, check for leakage, compare models, produce forecast diagnostics, and document failure cases.

Project 3 - Multi-Asset Allocation and Portfolio Construction

Research question: How should an investor allocate capital across major asset classes, and do more sophisticated portfolio methods improve on simple diversification?

A representative ETF universe may include U.S., developed international, and emerging-market equities; short- and long-term Treasuries; investment-grade and high-yield corporate bonds; emerging-market bonds; gold; commodities; and Bitcoin or a Bitcoin ETF.

Recommended analysis:

- Compare simple benchmarks such as equal weight or a traditional stock/bond portfolio with quantitative allocation methods.
- Implement methods such as minimum variance, mean-variance optimization, maximum Sharpe ratio, and risk parity.
- Study the sensitivity of optimized weights to expected-return and covariance estimates.
- Use realistic constraints such as no short sales, maximum weights, turnover limits, and transaction costs.
- Evaluate portfolio stability, drawdowns, rebalancing frequency, and out-of-sample performance.
- For advanced teams, consider Black-Litterman, covariance shrinkage, volatility targeting, or forecast-based tactical allocation.

Advanced extension: Regime-dependent asset allocation

More advanced teams may estimate macro or market regimes and allow expected returns, risks, correlations, or portfolio weights to vary across regimes. A possible architecture is: macro/market data -> regime classification -> regime-specific capital-market assumptions -> portfolio optimization -> risk controls. The regime model should be evaluated out of sample and compared with a non-regime benchmark.

AI opportunities: This track is especially well suited for an agentic design, for example a Data Agent, Capital-Market-Assumption Agent, Risk-Model Agent, Portfolio-Optimization Agent, Risk-Control Agent, and Reporting Agent. A fully autonomous system is not required; a carefully designed modular workflow is sufficient.

Project 4 - AI-Driven Equity Valuation and Industry Dashboard

Research question: Can traditional corporate valuation and industry analysis be rebuilt as a reproducible, code-based analytical application rather than a static Excel model?

Teams should select an industry and a manageable group of public firms, then construct a forecasting and valuation system using historical fundamentals, market data, and clearly documented assumptions.

Recommended analysis:

- Build revenue, margin, earnings, and free-cash-flow forecasts.
- Implement DCF and/or comparable-company valuation.
- Conduct scenario and sensitivity analysis for assumptions such as growth, margins, discount rates, terminal growth, and valuation multiples.
- Create an interactive dashboard using a framework such as Streamlit, Dash, Plotly, Shiny for Python, or another suitable platform.
- Explain which assumptions drive valuation most strongly and why point estimates should be accompanied by uncertainty analysis.

AI opportunities: Financial-statement extraction, data-pipeline construction, valuation-code generation, scenario engines, dashboard development, formula checking, automated diagnostics, and company/industry summaries.

Important requirement: The team should demonstrate that the dashboard is driven by transparent quantitative logic and reproducible code, not by an opaque AI-generated narrative.

Project 5 - Financial Text Analytics and Event Intelligence

Research question: Can information extracted from unstructured financial text help explain or predict returns, volatility, earnings outcomes, or corporate risk?

Possible text sources include 10-Ks, 10-Qs, earnings-call transcripts, company press releases, Federal Reserve communications, financial news, and analyst reports when legally available.

Recommended analysis:

- Create structured measures such as sentiment, uncertainty, readability, topic frequencies, semantic similarity, or changes in language over time.
- Compare traditional dictionary-based methods with modern NLP or LLM-based classification where appropriate.
- Link textual measures to financial outcomes such as abnormal returns, future volatility, earnings surprises, analyst revisions, or post-event drift.
- Use a clearly defined event window or prediction horizon and control carefully for timing to avoid look-ahead bias.
- Evaluate whether the textual signal adds incremental explanatory or predictive power beyond standard controls.

AI opportunities: Here AI can be both a coding tool and part of the empirical measurement process. Teams may use LLMs for classification, information extraction, summarization, embeddings, or document processing, but they must address prompt sensitivity, reproducibility, hallucination risk, model changes, and economic validity of the resulting measures.

7. Required AI Workflow Appendix

Every team must include an appendix that documents how AI was used throughout the project. This appendix is part of the graded submission, not an optional add-on.

- List the major AI tools used and what each tool was used for.
- Show representative prompts or instructions used for important tasks.
- Include screenshots or excerpts that illustrate the workflow, especially where AI performed coding, debugging, data engineering, testing, or workflow/agent design.
- Identify important AI-generated code or analysis that the team modified after testing or review.
- Document at least one case in which AI produced an incorrect, incomplete, or misleading output and explain how the team detected and corrected it.
- Explain how the team validated the final code and results. Examples include unit tests, independent calculations, benchmark comparisons, reasonableness checks, or replication against known results.
- If an agent architecture is used, include a diagram or description of the agents/modules, their inputs and outputs, and how control passes between them.

8. GitHub and Reproducibility Requirements

The GitHub repository is the official technical record of the project. It should be organized so that the instructor can inspect both the final outputs and the workflow that produced them. A typical structure includes the followings:

- README.md with project title, team members, research question, data sources, methodology summary, instructions for reproducing the analysis, and links to key outputs.
- Organized source-code folders rather than one very large script.
- A clear separation between raw data, processed data, code, figures/tables, and final outputs where feasible.
- A requirements.txt, environment.yml, pyproject.toml, or equivalent environment file when appropriate.
- Git commit history that shows meaningful development over time. Avoid uploading the entire project only once at the deadline.
- Final report and presentation files, or clearly labeled links to them.